Xiaoyue Mike Zheng

Contact Information

1 Bungtown Road
Cold Spring Harbor, NY 11724
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Education

Cold Spring Harbor Laboratory School of Biological Sciences, Cold Spring Harbor, NY

PhD Candidate, In Progress - Entered Fall 2020
Advisor: Arkarup Banerjee

Colby College, Waterville, ME

BA, May 2020
Major: Biology (Neuroscience Concentration) with Honor and Computer Science
Minor: Mathematics
Summa Cum Laude (GPA 4.16)

Honors and Awards

George A. and Marjorie H. Anderson Fellowship, CSHL School of Biological Sciences, 2023 - present

Honorable Mention, The Mathematical Contest in Modeling (MCM), 2017 & 2018

Colby College: Phi Beta Kappa, Phi Beta Kappa Achievement Award Scholar, Bixler Scholar, Strider Scholar, Presidential Scholar, 2016 - 2020

Academic Experience

Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

Graduate Student, School of Biological Sciences **Fall 2020 - present**

- Developed behavioral assays to quantitatively analyze the behavior and vocalizations of interacting rodents
- Investigated the role of the periaqueductal gray (PAG) in controlling vocalizations in singing mice

Advisor: Arkarup Banerjee

Icahn School of Medicine at Mount Sinai, New York, NY

Undergraduate Research Assistant, Department of Genetics and Genomic Sciences **Summer 2019**

- Developed BcForms, an open-source toolkit for concretely describing the molecular structure (atoms and bonds) of macromolecular complexes

Advisor: Jonathan Karr

Colby College, Waterville, ME

Research Assistant, Department of Computer Science **Spring 2019**

- Developed computational tools to fit circadian rhythm data and analyzed the effect of a small molecule on circadian rhythm

Advisor: Stephanie Taylor

Research Assistant, Department of Biology **2017 - 2020**

- Conducted a proteomics study to identify molecular pathways involved in a fruit fly model of Frontotemporal Dementia (FTD) and used genetic modifier screen to characterize the candidate genes

Advisor: Tariq Ahmad

Publications

Zheng, X.M.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2025). Vocal repertoire expansion in singing mice by co-opting a conserved midbrain circuit node. Current Biology 35, 5762-5778.e6.

Isko, E.C., Harpole, C.E., **Zheng, X.M.**, Zhan, H., Davis, M.B., Zador, A.M., and Banerjee, A. (2024). Selective expansion of motor cortical projections in the evolution of vocal novelty. Preprint at bioRxiv, https://doi.org/10.1101/2024.09.13.612752.

Lang, P.F.*, Chebaro, Y.*, **Zheng, X.***, P. Sekar, J.A., Shaikh, B., Natale, D.A., and Karr, J.R. (2020). BpForms and BcForms: a toolkit for concretely describing non-canonical polymers and complexes to facilitate global biochemical networks. Genome Biol 21, 117.

Lee, D.*, **Zheng, X.***, Shigemori, K., Krasniak, C., Bin Liu, J., Tang, C., Kavaler, J., and Ahmad, S.T. (2019). Expression of mutant CHMP2B linked to neurodegeneration in humans disrupts circadian rhythms in Drosophila. FASEB BioAdvances 1, 511–520.

Vandal, S.E., **Zheng, X.**, and Ahmad, S.T. (2018). Molecular Genetics of Frontotemporal Dementia Elucidated by Drosophila Models—Defects in Endosomal–Lysosomal Pathway. International Journal of Molecular Sciences 19, 1714.

* Equal contribution

Conference Talks

Zheng, X.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2024). Evolutionary diversification of vocal behavior requires the midbrain periaqueductal gray. Talk at 2024 Neural Mechanisms of Acoustic Communication GRS.

* Equal contribution

Select Poster Presentations

Zheng, X.M., Harpole, C.E., Davis, M.B., and Banerjee, A. (2025). Toward the neural code of a vocal innovation in the singing mouse. Poster presented at Neuroscience 2025 (SfN25).

Zheng, X.M.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2025). Neural basis of a vocal innovation in the singing mouse. Poster presented at the 2025 CNCM Conference (UC Irvine).

Zheng, X.M.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2024). Evolutionary diversification of vocal behavior requires the midbrain periaqueductal gray. Poster presented at Neuroscience 2024 (SfN24).

Zheng, X.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2024). Evolutionary diversification of vocal behavior requires the midbrain periaqueductal gray. Poster presented at 2024 Neural Mechanisms of Acoustic Communication GRC/GRS.

Zheng, X.*, Harpole, C.E.*, Vafidis, P., Davis, M.B., Cowley, B., and Banerjee, A. (2024). Distinct ancestral and novel social vocalizations in singing mice are controlled by the periaqueductal gray. Poster presented at Neuronal Circuits 2024 (CSHL).

Zheng, X.*, Harpole, C.E.*, Davis, M.B., and Banerjee, A. (2023). The role of the midbrain periaqueductal gray in controlling vocal flexibility in singing mice. Poster presented at Neuroscience 2023 (SfN23).

Zheng, X. and Ahmad, S.T. (2019). Proteomic analysis identified Echinoid as a genetic modifier of mutant CHMP2B associated with Frontotemporal Dementia. Poster presented at the 60th Annual Drosophila Research Conference.

Zheng, X. and Ahmad, S.T. (2018). Proteomic analysis of expression of mutant CHMP2B associated with Frontotemporal Dementia. Poster presented at the 2018 Maine Society for Neuroscience Annual Meeting.

Zheng, X., Heisler, A., and Ahmad, S.T. (2018). Proteomic analysis of expression of mutant CHMP2B associated with Frontotemporal Dementia. Poster presented at the 45th Maine Biological and Medical Sciences Symposium.

* Equal contribution

Teaching Experience

Mentor of two rotation students, CSHL, 2020 - present

Mentor of one undergraduate student (URP), CSHL, 2020 - present

Mentor of two high school students, CSHL, 2020 - present

Teaching Assistant, Department of Computer Science, Colby College, 2019

Teaching Assistant, Department of Mathematics and Statistics, Colby College, 2018

Tutor, Department of Computer Science, Colby College, 2018

Professional Experience

Organizer, Systems Neuroscience Journal Club, CSHL, 2022 - present

Computer Science (HPC) Intern, Bigelow Laboratory for Ocean Sciences, Summer 2020

Research Assistant, Maine Concussion Management Initiative, 2016 - 2020